CLOUD & DEVOPS

Helm Visual Guide

Charts, values, templates, releases — Kubernetes packaging done right.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Chart Anatomy

A chart is a versioned package of Kubernetes manifests.

Chart.yaml

name, version, appVersion

values.yaml

defaults the user can override

templates/

Go-templated manifests

charts/

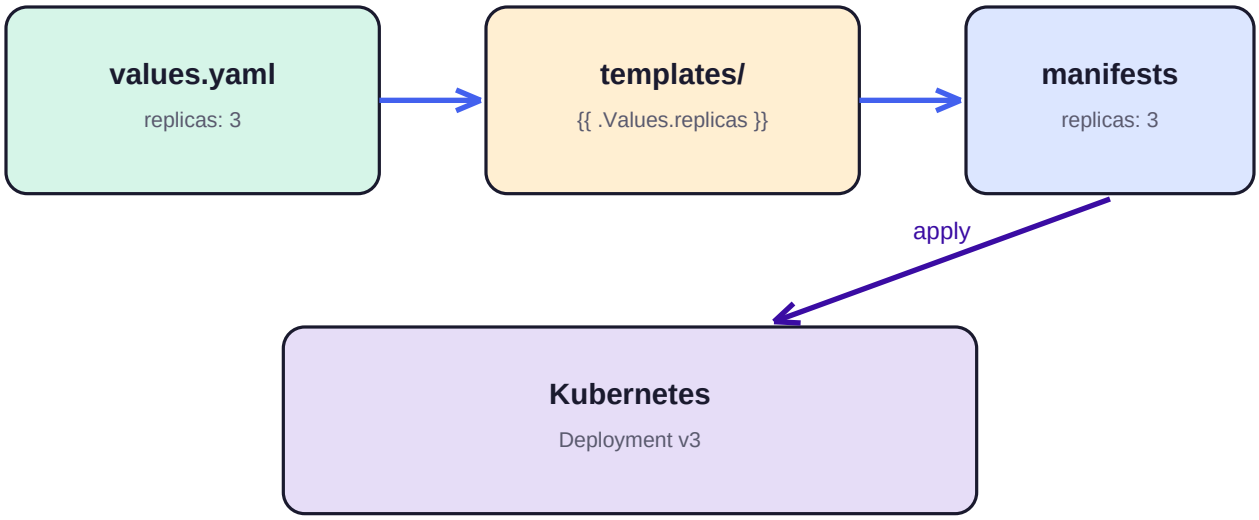
bundled sub-charts

Version BOTH chart and app: bump chart version on every change so upgrades are traceable.

STEP 2/6

Render Flow: Values → Templates → Manifests

Helm renders templates, then applies plain YAML to the cluster.



helm template renders locally — dry-run and diff before anything touches the cluster.

STEP 3/6

Release Lifecycle

Install → upgrade → rollback: every release is versioned.

helm upgrade --install: idempotent deploys — same command for first and tenth release.

- helm list shows every release and its revision number.
- helm history <release>: audit trail of what changed, when, by whom.
- One chart, many releases: dev/staging/prod differ only in values files.

STEP 4/6

Templating Power (and Traps)

Go templates make charts reusable — with discipline.

Conditionals

{{- if .Values.ingress.enabled }} — optional resources

Loops

{{- range .Values.hosts }} — N items from one block

Helpers

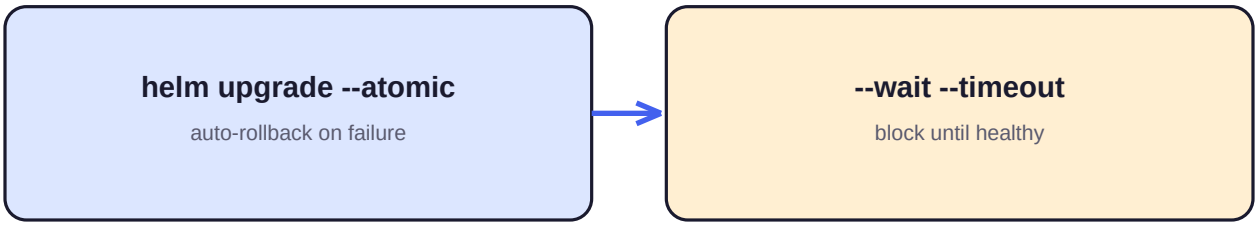
_helpers.tpl — shared labels/names, DRY charts

Trap: templates are string surgery — always `helm template | kubectl apply --dry-run=client` after edits.

STEP 5/6

Upgrades & Rollbacks in Practice

Atomic upgrades fail fast; rollbacks restore the last good state.



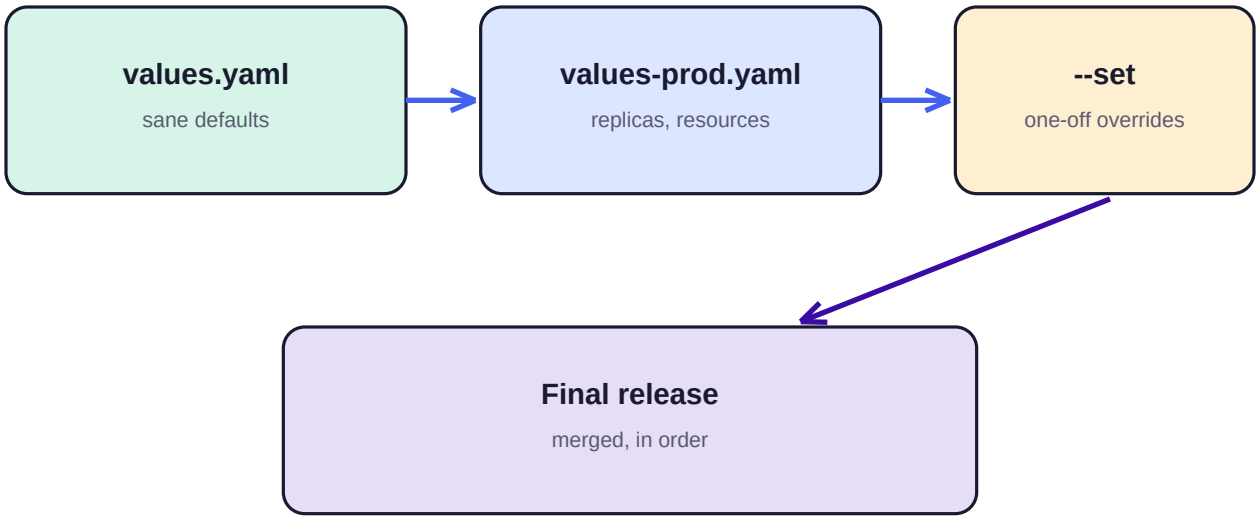
A broken v2 never becomes the current release — traffic stays on v1.

- --atomic: if the upgrade fails, Helm rolls back to the previous revision automatically.
- --wait: don't declare victory until pods are actually Ready.
- helm rollback <release> [revision]: one command back to safety — practice it in staging.

STEP 6/6

values.yaml Strategy

Layered values keep environments honest.



Later files win. Keep secrets OUT of values — use sealed-secrets or external-secrets instead.

STEP ✓/✓

Key Takeaways

Helm — Remember

- A chart = Chart.yaml + values.yaml + templates: versioned, reusable K8s packaging.
- Render flow: values fill templates → plain manifests → applied to the cluster.
- helm upgrade --install --atomic --wait: the safe, idempotent deploy command.
- Layer values (defaults → env file → --set); never put secrets in values.
- history + rollback make every release reversible — practice rollbacks.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev